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**GCNFormNet: Branched Graph-Transformer Architecture for
EEG-Based Emotion Recognition**

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Abstract

Existing EEG-based emotion recognition pipelines depend heavily on complex, multi-stage preprocessing and hand-crafted feature extraction, making it difficult to judge how much of a model's reported performance actually comes from its architecture. To address this, Raghav and Indolia (2026) propose GCNFormNet, a dual-branch hybrid architecture that learns spatio-temporal EEG representations directly from minimally processed signals. One branch is a Graph Convolutional Network (GCN) that models spatial relationships between EEG channels using a dynamically generated, time-varying adjacency matrix, removing the need for a fixed, anatomically predefined electrode graph. The parallel branch is a Transformer that captures long-range temporal dependencies using Performer-based linear self-attention, with standard Layer Normalization replaced by DynamicTanh (DyT), an element-wise, normalization-free activation. The two feature vectors are concatenated and passed to a dense classifier for the final emotion label.

The model is evaluated within the standardized EEGain framework on four benchmark datasets — DEAP, DREAMER, SEED, and SEED-IV — using the strict Leave-One-Subject-Out (LOSO) protocol, which better reflects real-world, subject-independent generalization than commonly used subject-dependent splits. GCNFormNet achieves the best or near-best accuracy on the categorical datasets, notably the highest accuracy among compared methods on SEED-IV (0.46), supported by Welch's t-tests. Performance on the binary valence/arousal datasets (DEAP, DREAMER) is competitive but not clearly superior, and a trivial majority-class baseline is sometimes hard to beat, highlighting the difficulty of cross-subject generalization.

An interpretability analysis of the learned adjacency matrices reveals neurophysiologically plausible patterns discovered purely from data, without anatomical priors — including hemispheric asymmetry linked to valence on DREAMER, prefrontal dominance for negative/high-arousal emotions on SEED and SEED-IV, and posterior-occipital engagement for high valence on DEAP. An ablation study confirms the Transformer branch forms the performance backbone while the GCN branch contributes complementary gains that scale with electrode density, being most useful on the 62-channel SEED/SEED-IV datasets and mildly counterproductive on the lower-density 32-channel DEAP dataset. Sensitivity analysis of the adjacency-matrix hyperparameters and DyT vs. LayerNorm further validate the design. Overall, GCNFormNet shows an end-to-end, minimally preprocessed pipeline can match or exceed feature-engineered baselines on multi-class EEG emotion recognition.

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1. Introduction

Recognizing human emotion is a core problem in affective computing, underpinning applications such as adaptive e-learning, empathetic healthcare robots, and real-time mental health monitoring. Behavioral cues like facial expression and speech are susceptible to conscious masking and environmental noise, whereas physiological signals offer a more objective, tamper-proof window into internal emotional states. Among these, electroencephalography (EEG) is especially attractive because it captures the real-time neural processes underlying emotion with high temporal resolution, and increasingly affordable, portable recording hardware makes it practical for real-world brain-computer interface (BCI) applications.

Deep learning has driven major progress in EEG-based emotion recognition, moving from CNN-based models such as EEGNet toward architectures that integrate Transformers for long-range temporal modeling. However, a persistent gap remains between the end-to-end philosophy of deep learning and its practical implementation: many state-of-the-art models still rely on complex, multi-stage preprocessing and hand-crafted features such as Differential Entropy (DE) or Power Spectral Density (PSD). This makes it hard to tell whether reported performance gains come from the architecture itself or from meticulous feature engineering, which in turn limits genuine cross-subject generalization.

To address this, the authors propose GCNFormNet, a dual-branch architecture designed to learn spatio-temporal features directly from minimally processed EEG signals. A Graph Convolutional Network (GCN) branch models spatial relationships between electrodes through a dynamically generated adjacency matrix, while a parallel Transformer branch captures complex temporal dynamics using Performer-based self-attention, with Layer Normalization replaced by DynamicTanh (DyT). The model is evaluated on four benchmark datasets within the standardized EEGain framework using the rigorous Leave-One-Subject-Out (LOSO) protocol, and an ablation study verifies the synergistic contribution of the two branches.

2. Literature Review

Deep learning approaches to EEG emotion recognition began with CNN-based methods, most notably EEGNet, which used depthwise and separable convolutions tailored to EEG signals, and ConvNet variants (Shallow/Deep) that served as robust baselines. More recent work has integrated Transformer-based architectures, including hybrid CNN-Transformer models such as MSDCGNet, to combine local feature extraction with global temporal dependency modeling.

Preprocessing methodology varies widely across the field. EEGain, adopted in this paper, uses minimal, standardized preprocessing (resampling and simple filtering) for fair comparability. In contrast, many published models rely on hand-crafted features such as Differential Entropy (DE), PSD, or wavelet transforms, often combined with class-balancing (e.g., SMOTE-Tomek) and PCA-based dimensionality reduction. Representative examples include DDNet, CA-ResNet-DenseNet, TPRO-Net, Emotion Perceptor, MSC-Attention-LSTM, and Hybrid-GRU-RNN, each reporting high accuracy under feature-engineered, often subject-dependent protocols.

The EEGain framework exposed major inconsistencies in the field: most studies validate on only one or two datasets, use inconsistent preprocessing and label-thresholding conventions, and report widely different accuracies depending on whether Leave-One-Trial-Out (LOTO, ~80–95% accuracy) or the far stricter Leave-One-Subject-Out (LOSO, ~50–60% accuracy) protocol is used. This motivates GCNFormNet's strategy of prioritizing LOSO for realistic subject-independent evaluation, with a standardized, feature-light preprocessing pipeline to isolate the architecture's true contribution.

3. Methodology

3.1 Overall Architecture

GCNFormNet is a hybrid model with two parallel branches operating on the same temporally convolved feature map: a GCN branch for spatial feature extraction and a Transformer branch for temporal feature extraction. The input EEG signal $X \in \mathbb{R}^{(B \times 1 \times C \times T)}$ (batch, 1 channel, C electrodes, T time steps) first passes through a temporal convolution (kernel size 25) to produce a shared feature map, which is then routed into the two branches. Their outputs are concatenated and passed to a dense classifier that outputs the final emotion class probabilities, as illustrated in Fig. 1.

3.2 Graph Branch (Spatial Extraction)

The GCN branch generates a time-step-wise dynamic adjacency matrix A_t for each sample, computed from learnable query/key projections of the channel features H_t , scaled by a temperature parameter τ , passed through Softmax, weighted by a learnable sparsity gate $\sigma(s)$, and combined with an identity matrix I for self-loops:

$$A_t = \sigma(s) \cdot \text{Softmax}((H_t W_q)(H_t W_k)^T / \tau) + I \quad (1)$$

This matrix drives three stacked GCN layers (hidden dimension 64) with ELU activations. For layer l , the update rule is:

$$Ht(l+1) = ELU(At \cdot Htl \cdot Wgl) \quad (2)$$

The final spatial vector Z_g is obtained by global average pooling of the last GCN layer's output across channels and time. Since the adjacency matrix is learned rather than fixed by electrode geometry, the model discovers context-specific inter-channel relationships purely from data.

3.3 Transformer Branch (Temporal Feature Learning)

The Transformer branch applies a spatial (channel-mixing) convolution to the shared feature map to obtain X_s , projects it to an embedding dimension of 128, and adds learnable positional embeddings P_{pos} :

$$X_{emb} = Proj(X_s) + P_{pos} \quad (3)$$

The resulting sequence passes through two Temporal Transformer Blocks, each combining Performer-based linear self-attention with a dilated 1D convolution to capture global and local temporal patterns, and using DynamicTanh (DyT) in place of Layer Normalization:

$$X' = DyT_1(X_{in} + Performer(X_{in}) + DilatedConv(X_{in})) \quad (4a)$$

$$H_{out} = DyT_2(X' + FFN(X')) \quad (4b)$$

The final temporal vector Z_t is obtained via global average pooling of H_{out} over time.

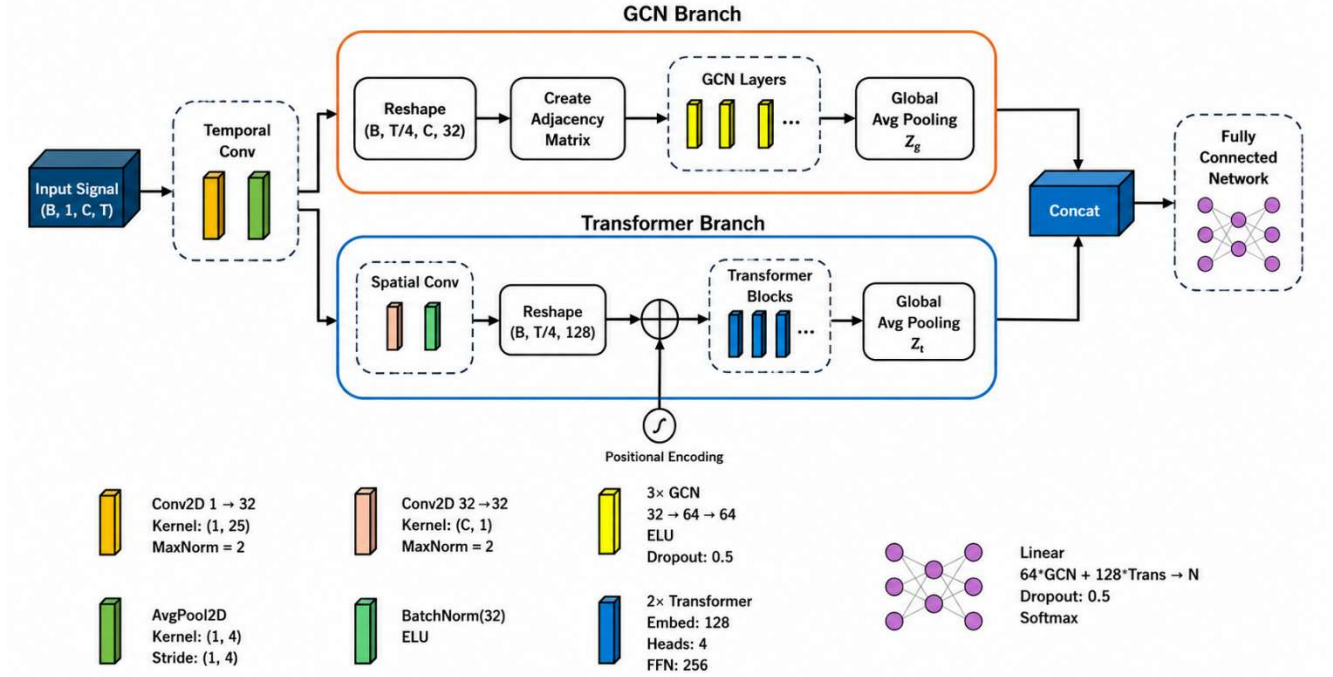


Fig. 1. Architecture overview of GCNFormNet.

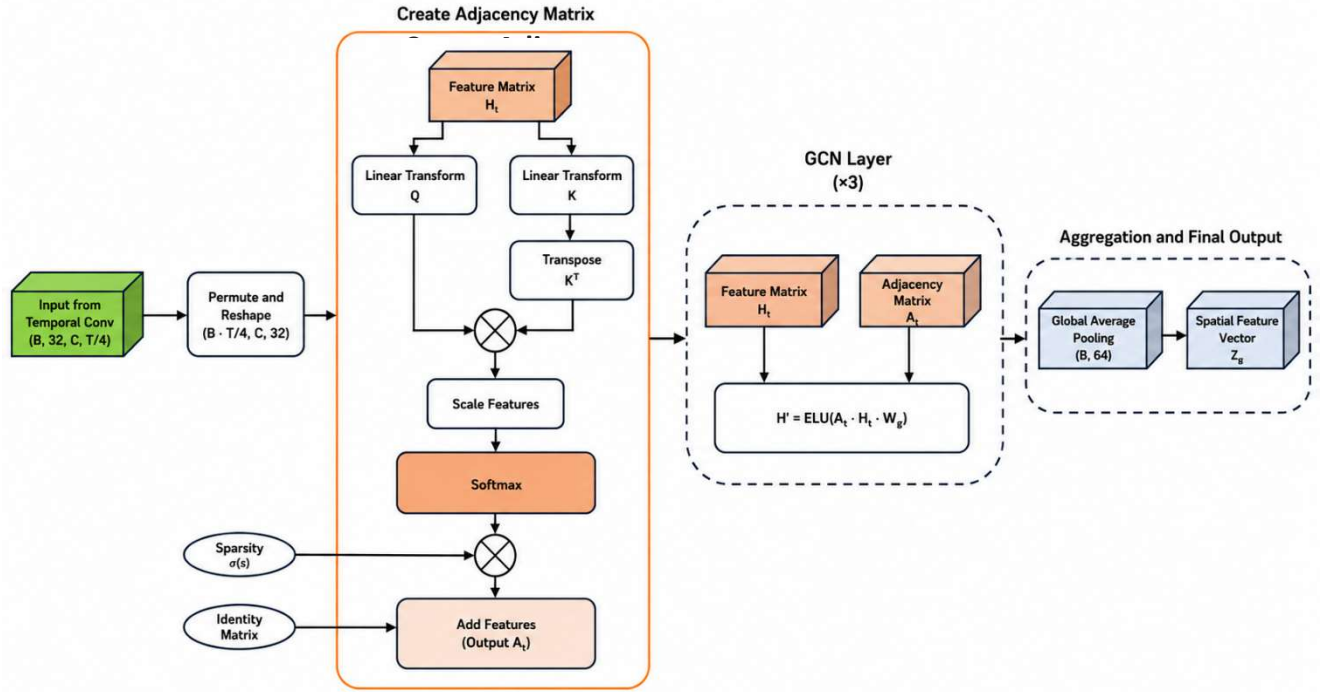


Fig. 2. GCN Branch of GCNFormNet.

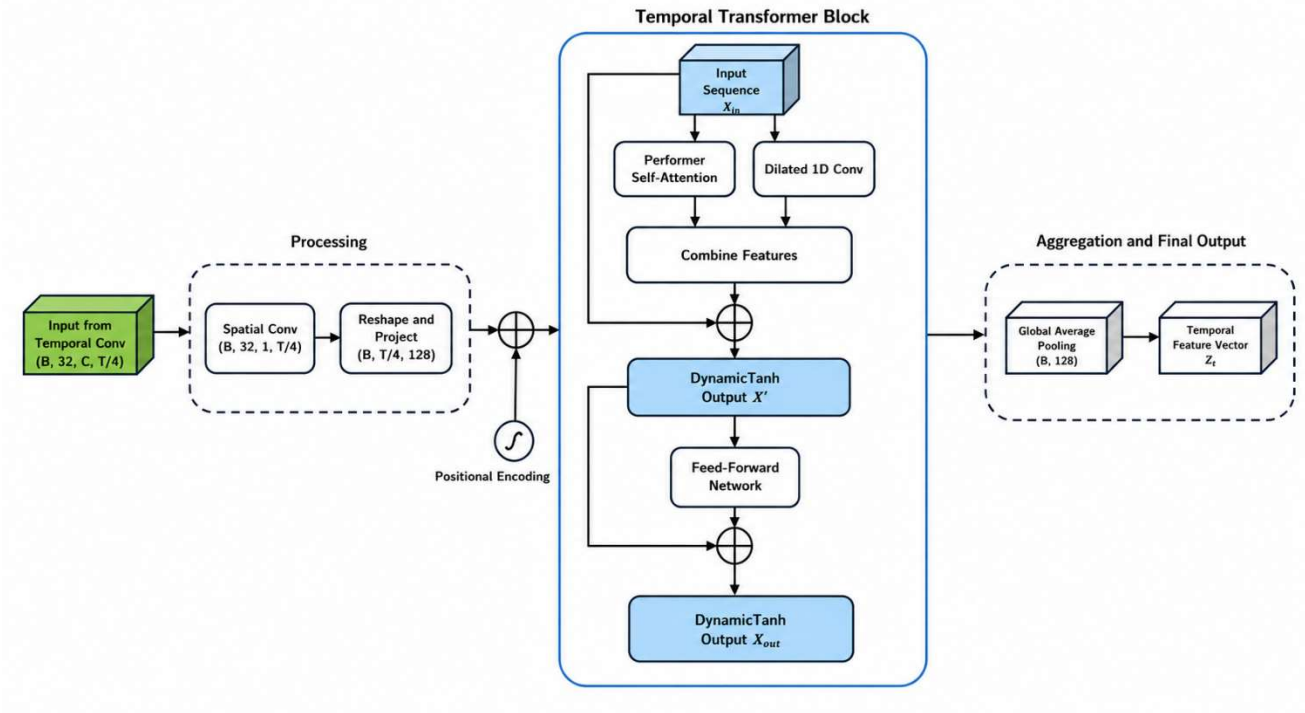


Fig. 3 Transformer Branch of GCNFormNet

3.4 Feature Fusion, Datasets, and Evaluation Protocol

Z_g and Z_t are concatenated, passed through dropout, and fed to a final softmax classifier to produce the emotion logits $\hat{y}$:

$$Z_{comb} = [Z_g ; Z_t], \hat{y} = \text{Softmax}(W_c \cdot \text{Dropout}(Z_{comb}) + bc) \quad (5)$$

The model is evaluated on four EEGain-integrated datasets: DEAP (32 subjects, 32-channel, valence/arousal), DREAMER (23 subjects, 14-channel dry electrodes, valence/arousal), SEED (15 subjects, 62-channel, 3-class), and SEED-IV (same subjects, 62-channel, 4-class). All experiments use Leave-One-Subject-Out (LOSO), training for 200 epochs with Adam (lr 0.001), label smoothing (0.01), and dropout (0.5). Baselines include EEGNet, ShallowConvNet, DeepConvNet, TSception, a Trivial Baseline, and MSC-Attention-LSTM, assessed via Accuracy, F1, and Weighted F1.

4. Results and Discussion

GCNFormNet achieves its strongest results on the categorical, multi-class datasets. On SEED-IV, it obtains the highest accuracy among all compared models (0.46 ± 0.08), significantly outperforming EEGNet ($p < 0.001$), ShallowConvNet ($p < 0.01$), TSception ($p < 0.05$), and the Trivial Baseline ($p < 0.001$). On SEED, it reaches 0.54 ± 0.07 accuracy, statistically comparable to the best baseline (DeepConvNet, 0.55 ± 0.08) but with lower variance, while significantly beating TSception, EEGNet, and the Trivial Baseline. Table 1 summarizes the headline LOSO results across all four datasets.

Dataset	Task	GCNFormNet Accuracy	Best Baseline (Accuracy)
DEAP	Valence	0.50 ± 0.09	Trivial Baseline: 0.57 ± 0.09
DEAP	Arousal	0.52 ± 0.11	Trivial Baseline: 0.59 ± 0.15
DREAMER	Valence	0.57 ± 0.07	MSC-Attention-LSTM: 0.64 ± 0.05
DREAMER	Arousal	0.48 ± 0.08	Trivial Baseline: 0.52 ± 0.15
SEED	3-class	0.54 ± 0.07 (best)	DeepConvNet: 0.55 ± 0.08
SEED-IV	4-class	0.46 ± 0.08 (best)	DeepConvNet: 0.45 ± 0.09

On the binary tasks, results are more mixed. On DEAP, cross-subject generalization is difficult for every model, and the Trivial Baseline significantly outperforms GCNFormNet on valence accuracy ($p < 0.01$) and weighted F1 ($p < 0.001$) — showing that complex architectures can struggle to beat the majority-class prior under strict LOSO. On DREAMER, MSC-Attention-LSTM's recurrent design achieves the

best valence accuracy (0.64 ± 0.05), significantly outperforming GCNFormNet ($p < 0.001$), while arousal differences are largely non-significant.

A t-SNE interpretability analysis shows GCNFormNet's learned features are far more separable by class than raw EEG features across all datasets, confirming the architecture learns discriminative structure from minimally processed signals. The learned adjacency matrices further reveal neurophysiologically meaningful patterns discovered without anatomical priors: hemispheric asymmetry linked to valence on DREAMER, stronger temporal-occipital coupling for high valence on DEAP, and prefrontal-dominant connectivity distinguishing emotional engagement from neutral baseline activity on SEED and SEED-IV.

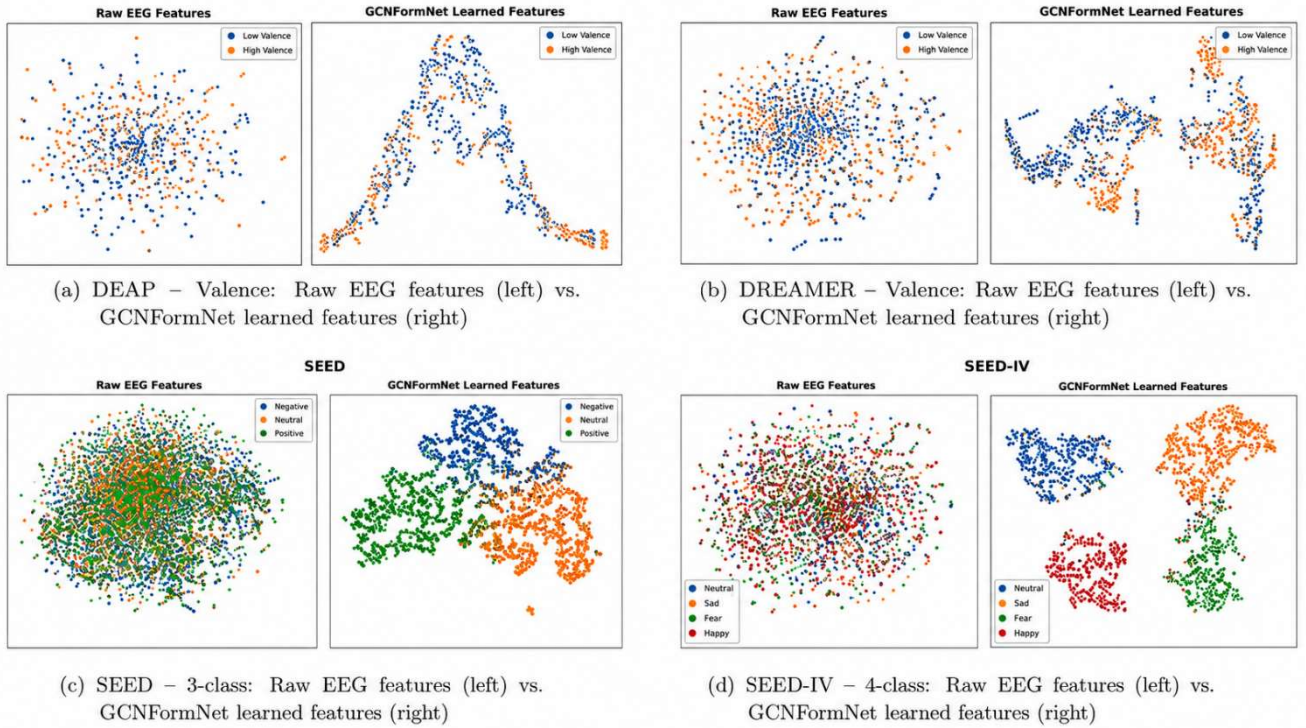


Fig. 4. t-SNE projections comparing raw EEG features with GCNFormNet's learned features (right) across DEAP, DREAMER, SEED, and SEED-IV, showing markedly improved class separability (Raghav and Indolia 2026).

The ablation study shows the Transformer branch consistently forms the performance backbone, while the GCN-only variant is weakest across all datasets. The full hybrid model performs best on DREAMER, SEED, and SEED-IV, confirming genuine synergy. The exception is DEAP, where the GCN branch overfits to subject-specific spatial patterns on the lower-density 32-channel montage that do not transfer to held-out subjects — its benefit scales with electrode density, being most valuable on the 62-channel

SEED/SEED-IV datasets. Sensitivity analysis confirms the default hyperparameters ($\tau = 0.1$, sparsity init = 0.5) are near-optimal, and DyT matches or exceeds LayerNorm in both accuracy and throughput.

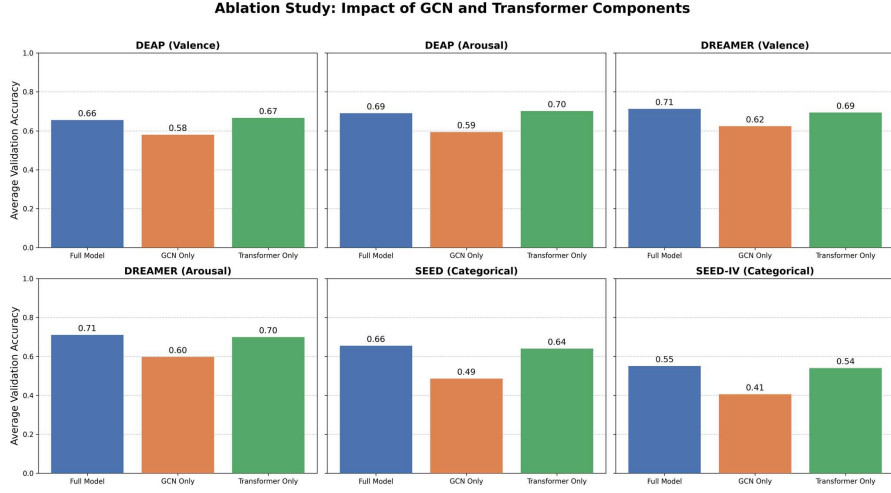


Fig. 5. Average validation accuracy of the ablation study (Full Model vs. GCN-only vs. Transformer-only) across all six dataset-task combinations (Raghav and Indolia 2026).

5. Limitations and Future Research

The authors identify two key limitations. First, the strong performance of the Trivial Baseline on DEAP and DREAMER suggests that on datasets with high inter-subject variability or class imbalance, complex architectures can struggle to outperform the simple majority-class prior, pointing to a need for models with greater inherent robustness to these data characteristics. Second, significant inter-subject variability remains even with GCNFormNet's more stable (lower-variance) predictions, especially on SEED-IV and DEAP, indicating that minimizing preprocessing alone is not sufficient to fully overcome subject-specific neural differences.

Proposed future directions include adaptive branch weighting through gating or attention mechanisms to resolve the observed dataset-specific performance variation between the GCN and Transformer branches, along with extending evaluation to all six datasets supported by the EEGain framework for a more comprehensive assessment of generalization.

6. Conclusion

GCNFormNet is a hybrid, dual-branch architecture that combines a dynamically-graphed GCN with a Performer-based Transformer to learn spatio-temporal EEG representations directly from minimally processed signals, avoiding the heavy feature engineering common in prior work. Evaluated under the strict, subject-independent LOSO protocol across four benchmark datasets, it achieves the best or statistically comparable performance on the more challenging multi-class categorical tasks (SEED, SEED-IV), while remaining competitive but not clearly superior on the binary valence/arousal tasks (DEAP, DREAMER). Its learned adjacency matrices surface neurophysiologically plausible connectivity patterns — such as hemispheric asymmetry and prefrontal dominance — purely from data, and its ablation study confirms genuine synergy between the spatial and temporal branches on higher-density electrode datasets. Overall, the work demonstrates that a carefully designed end-to-end architecture can match or exceed feature-engineered baselines, while also highlighting open challenges in cross-subject generalization that motivate future work on adaptive branch weighting and broader dataset validation.

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